· M E G A N S M I T H ·

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Expert Data Analyst with a strong background in healthcare analytics, data analysis, visualization, and interpretation, specializing in HEDIS quality measures and reporting. Skilled in integrating complex datasets and delivering audit-ready insights to diverse stakeholders by identifying key trends in healthcare performance. Experienced in SQL, Tableau, Python, R, and Excel, with expertise in data validation, root-cause analysis, and dashboard development. Adept in translating complex data into actionable recommendations that guide strategic decision making.

SKILLS

Data Analysis: SQL | Tableau | Python | R | Excel
Data Visualization: Tableau | Excel | Python
Programming Languages: SQL | Python | R
Statistical Analysis & Machine Learning:
Multivariate Regression | Time Series Analysis | Anomaly Detection
Predictive Modelling | Correlation Analysis

Certifications & Leadership Training:
Explore Leadership Series – BCBSM
Data Analytics Certification – Data Career Jumpstart
Recognition: EMMY Awards for Efficiency (2) and Integrity (1)
Other Programs: Microsoft Azure | Databricks
PowerPoint | Word | Canva | Google Suite

WORK EXPERIENCE

EXPERT ANALYST, HEDIS STRATEGIC IMPROVEMENT
Emergent Holdings *11/2024 – Present*

- Synthesized complex supplemental and claims data across 5 Medicare health plan contracts, delivering audit-ready insights and accurate benchmarking and performance analysis for 15+ HEDIS (Healthcare Effectiveness Data and Information Set) administrative and hybrid measures.
- Leveraged Microsoft Azure, Databricks, and Python to validate and analyze large-scale datasets, identifying data discrepancies and improving overall data quality and reliability for downstream reporting.
- Designed and implemented interactive dashboards and reporting using Excel and PowerPoint for 9+ HEDIS and Stars measures across 3 health plans, enabling leadership to identify trends and drive targeted interventions that improved measure performance.
- Created and presented provider organization dashboards to key stakeholders across 5 Medicare contracts, driving engagement and effectively communicating key performance forces, success areas, and strategic opportunities for improvement.
- Defined, calculated, and validated key performance metrics and trend analyses, including year-over-year comparisons, to identify drivers of measure performance and uncover opportunities for quality improvement and gap closure.
- Partnered with cross-functional stakeholders and external vendors to translate business requirements into analytical solutions, delivering executive-level insights and actionable findings that supported strategic decision-making and performance improvement initiatives.

MATHEMATICS EDUCATOR
Rochester Community Schools *08/2010 – 11/2024*

- Engineered Excel-based performance spreadsheets to track student achievement metrics and assessment outcomes, transforming raw data into actionable insights that supported targeted interventions and drove a 5% increase in overall student performance.
- Analyzed qualitative and quantitative student scores to determine standards and objectives that needed remediation, resulting in significant curriculum adjustments that upgraded the quality of 200+ instructional materials for cross-functional teams and faculty.
- Enhanced learning by defining and articulating learning outcomes, course goals, and test scores and analyzing performance metrics and measurements of over 1,400+ students.
- Documented upfront and ongoing needs assessments for over 80+ targeted learners, including gap analysis, research, and evaluation to ensure 7+ courses matched curricula standards and identified opportunities for improvement.

EDUCATION

Master of Science in Industrial Applied Mathematics, *summa cum laude requirements met* | *Oakland University*
Bachelor of Science in Secondary Mathematics & Biology Education, *magna cum laude* | *Rochester Christian University*

PROJECTS

SQL: HEALTHCARE DATA

- Utilized SQL to examine hospital patient data spanning 10 years and covering over 101,000 diabetic patients, identifying correlations between medical specialties and patient procedures and assessing the impact of race on medical treatments.
- Analyzed healthcare data to reveal equitable treatment for diabetic patients regardless of race, highlighting an inclusive approach and commitment to unbiased healthcare services.
- Translated analytical findings into actionable recommendations, including recommendations for optimizing hospital operations, enhancing patient care, and leveraging success stories for public relations initiatives.

TABLEAU: EDUCATION DATA

- Conducted school performance analysis to evaluate graduation rates and demographic data of 1,800+ schools, distinguishing top and bottom performers based on graduation statistics, economically disadvantaged students, and students with high needs.
- Investigated geographic patterns of educational outcomes using Tableau, mapping graduation rates to school locations and identifying clusters of high and low-performing schools, informing allocation and policy initiatives.
- Orchestrated the development of an interactive Tableau dashboard, presenting actionable insights to an analyst team at Kaplan Inc., proposing strategies to improve student support and enhance academic success.

PYTHON: MANUFACTURING DATA

- Interpreted manufacturing time series data containing 730,000 samples, investigating anomalies and identifying correlations between critical variables impacting plant performance.
- Applied statistical techniques using Python to identify linear correlations and generated data visualizations confirming the stability of critical variables.
- Developed Python visualizations to investigate fluctuations in material purity during manufacturing, indicating seamless operations and a 10% increase in desired material post-processing.